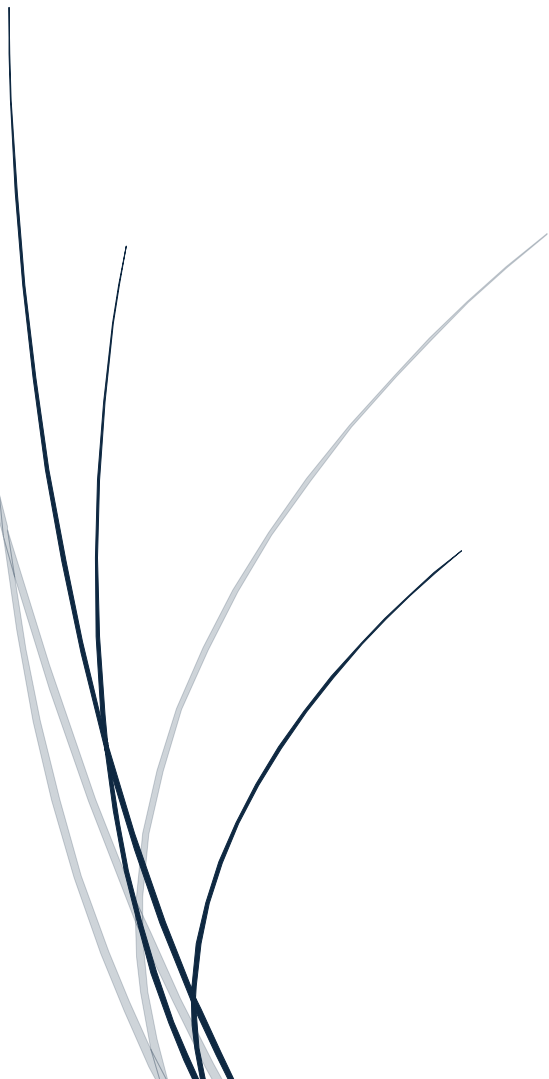


Research Proposal & PRD- designing

Names: Alberto Barrionuevo
Pantelis Paraskevopoulos
Jasper Naalden
Nasjaat Potmis

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HAN_UNIVERSITY
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Version Control

Version	Addition
0.1	First draft
0.2	Second draft
0.3	Third draft

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1. AIBS – Research Proposal (first draft)

1.1 Target Audience

This research focuses on Dutch mechanical engineering and industrial product design firms with 20–75 employees that use CAD software such as SolidWorks, Autodesk Inventor, and Siemens NX. These companies want to determine whether AI can improve their design pipeline by reducing design time, improving design quality, and automating repetitive engineering tasks, while ensuring that the benefits outweigh the costs of implementation.

However, the growing integration of AI into 3D design software may increase dependence on a small number of mainly U.S.-based software providers, as many advanced AI features are developed by the software companies themselves. To reduce this dependency and strengthen strategic autonomy, organizations should explore alternative AI models, open-source solutions, and European or other non-American technology ecosystems. Diversifying AI providers can improve resilience, reduce supplier dependence, and increase long-term flexibility.

1.2 What do these companies need?

We see that companies are too afraid to start using AI in their business. Because they think the way of working now is perfect for them. There is also a lack of knowledge about the things AI can do, by the people who make these decisions in these businesses. But they are forced to change so they need to decide are we going to transform our company and be a more future realistic company, or fall behind and let other rival companies capitalize in their market.

According to Draghi the innovation gap between the US/China and Europe is big so companies in Europe need to decide if they want to close that gap or not. Because it's in their own hands if companies want to be more successful according to Draghi they need to increase their investment more in AI, digital technologies, research and innovation. Instead of focussing on the present and results.

1.3 Questions & Methodology

1.3.1 Main question

How can designers and engineers use AI to improve their design pipeline?

1.3.2 Sub questions

Sub-question 1: What does a design engineer do on a daily bases?	
Sub-question type	Field
Goal	Current work for a design engineer
Research-strategy	Document analysis - Interview

Sub-question 2: What AI tools do exist for a Designers and Engineer?	
Sub-question type	Desk & Field
Goal	What is already available and how dan this be used.
Research-strategy	Document analysis - Interview

Sub-question 3: What parts of the designing can be supported or automated by AI?	
Sub-question type	Desk & Field
Goal	Insight in the possibilities of AI to support or automate task for design-engineers
Research-strategy	Document analysis - Interview

Sub-question 4: What do SME decision-makers currently know about AI applications for design engineering, and where are the gaps?	
Sub-question type	Desk & Field
Goal	Insight in the possibilities of AI to support or automate task for design-engineers
Research-strategy	Document analysis – Interview/Questionnaire

Sub-question 5: Has any comparable mechanical-engineering or product-design SME already implemented AI in their design pipeline, and what were the results?	
Sub-question type	Desk & Field
Goal	Identify a real case where AI has already been used in a comparable design pipeline, to compare against perceived opportunities and risks.
Research-strategy	Document analysis (case studies, industry press) - Interview

Sub-question 6: What opportunities and risks brings AI for a designers and Engineers?	
Sub-question type	Desk & Field
Research-strategy	Document analysis - Interview
Goal	Get insight in how design-engineers see the opportunities and risks in the use of AI in their work

1.4 Quality criteria for the handbook

- **Understandable**
 - Engineers and designers must be able to understand the content without needing AI expertise.
 - Technical terms are explained when necessary.
- **Practically applicable**
 - The handbook provides recommendations and examples that can be directly applied in daily design work.
- **Balanced**
 - Both opportunities and risks of AI are discussed, allowing readers to make informed decisions.
- **Action-oriented**
 - Readers understand what actions they can take to start using AI in their work.
- **Up-to-date**
 - The handbook includes current AI tools, technologies, and developments.
- **Well-structured**
 - Information is logically organized and easy to navigate.
- **Easy to use**
 - Readers can quickly find the information they need through clear headings, page numbers, and a table of contents.
- **Evidence-based**
 - Recommendations are supported by research findings and interview results.
 - This is reinforced by Sub-question 5, which compares design-engineers' perceived opportunities and risks against a documented case of a comparable SME that has already implemented AI in its design pipeline.

1.5 Search strategies

The strategies below extend AIBS 1.3.2 (sub-questions) and PRD 2.4 (known constraints on literature). Each sub-question already names a research strategy (document analysis, interview, or both); this section turns "document analysis" into a query the agent can run, per type of source.

1.5.1 Desk research: academic and professional literature

Applies to sub-questions 1, 2, 3, 4 and 6 (AIBS 1.3.2).

Item	Detail
Databases / sites	CBS Statline, Kamer van Koophandel (KVK) publications, sector organisations (e.g. FME, Koninklijke Metaalunie) Google Scholar, ResearchGate, IEEE Xplore Vendor documentation: SolidWorks, Autodesk Inventor, Siemens NX EU and national programme sites (NL AI Coalition, EU Digital Decade)
Query template	("artificial intelligence" OR AI OR "generative design") AND ("mechanical engineering" OR "product design" OR CAD) AND (SME OR "midsize manufacturer" OR MKB)
Dutch equivalent	("kunstmatige intelligentie" OF AI) EN ("werktuigbouwkunde" OF productontwerp OF CAD) EN (MKB OF "midden- en kleinbedrijf")
Filters the agent applies	Publication date: within the last 4 years of the search date (PRD 2.4) Language: Dutch or English only (PRD 2.4) Format: PDF or HTML from a publisher, journal, government or vendor domain

1.5.2 Sub-question 5: comparable case studies

Applies to sub-question 5 (has a comparable SME already implemented AI in its design pipeline?).

Item	Detail
Sources	Trade press (e.g. Link Magazine, Engineering.com, sector news) Case-study libraries published by CAD vendors Conference proceedings and white papers from professional bodies
Query template	("case study" OR "pilot project") AND (AI OR "generative design") AND ("CAD" OR "mechanical engineering" OR "product design") AND (implementation OR adoption OR results)
Extra filter	Company size stated or inferable as comparable to the 20–75 employee target range (AIBS 1.1); if size cannot be established, the source is flagged for human review rather than accepted or rejected automatically.

1.5.3 Field material: interview transcripts

Interview transcripts are not gathered through the search agent at all. They are supplied directly by a team member (PRD 2.2, step 1) and follow the separate handling rules in PRD 2.4 ("Handling of interview material"): raw, unredacted transcripts never enter the automated search or quality pipeline described below, and never reach a model service until a team member has redacted them (see blueprint 1.4, "The line nothing may cross").

1.6 Source quality criteria

The agent applies the checklist below to every source found through section 1 before it is treated as usable. The first four criteria are objective enough to check automatically; relevance is tagged by the agent but confirmed by a person, consistent with the two-layer check the platform already uses elsewhere (blueprint 1.2.2, Decision 6).

Criterion	Rule	Agent action
Recency	No more than 4 years old (PRD 2.4).	Auto-reject if older; returns to Collect with reason.
Source type	Academic, professional, government or a recognised vendor/knowledge organisation. No Wikipedia or blogs without a named author (PRD 2.4).	Auto-reject if source_type cannot be classified into an accepted type.
Language	Dutch or English only (PRD 2.4).	Auto-reject otherwise.

Publisher reputation	Recognised publisher, professional journal, government body or reputable knowledge organisation (PRD 2.4): e.g. CBS, KVK, EU institutions, NL AI Coalition, established trade press.	Checked against a maintained allow-list; unlisted publishers are flagged, not rejected, for a person to decide.
Traceability	Enough metadata (author/organisation, title, date, publisher) to write an APA 7 reference (AIBS 1.4, "include citations").	Flagged if metadata is incomplete; not auto-rejected, since metadata can sometimes be completed by hand.
Relevance	Content addresses at least one of the six AIBS sub-questions (AIBS 1.3.2) directly.	Agent tags the source with the sub-question(s) it addresses; a person confirms the tag before the source is used in a chapter.

1.6.2 Building this into the tool

These criteria are not a separate step bolted onto the pipeline: they map directly onto the appraised.md contract already defined in blueprint 1.3.

- Recency, source type and language are exactly the source_type, publication_date and language fields Appraise fills in for every source (blueprint 1.3). The automatic PRD 2.4 check the Appraise role already performs (blueprint 1.2.2) is what enforces these three criteria mechanically.
- Publisher reputation is added as a named allow-list that the agent checks source_type and publisher against, rather than leaving "recognised publisher" to judgement alone.
- Relevance tagging is added as an explicit output of Appraise's model layer, alongside the summary, key_findings, opportunities and risks fields it already writes (blueprint 1.3).
- The search strategies in section 1 become the query templates the Collect role runs when a team member asks it to gather sources for a given sub-question. Each template is stored as an editable instruction, not code, so it can be updated as a search proves too broad or too narrow, in line with the "agentic CLI" design consequence in PRD 2.4: behaviour changes by editing an instruction rather than rewriting code.
- A source that fails an automatic criterion is returned to Collect with the reason attached, exactly as blueprint 1.2.1 already specifies for a failed Appraise check; the default stays "stop, not publish" (blueprint 1.5).

1.7 Company interview

Companies that exist and we could interview

Name	Location	What do they do?	Why are they interesting for us and how are they using a design engineer?
CAD2M	Doetinchem	Works with SOLIDWORKS, CAD/CAM, PDM, 3D design and engineering software.	Very relevant because they work directly with CAD and are also interested in AI and digitalisation in engineering. Create and modify 3D CAD models, support customers with SOLIDWORKS, CAD/CAM and PDM software, improve digital design workflows, and advise companies on new technologies such as AI in engineering.
VIRO	Arnhem	Engineering company working on mechanical design, machine building and industrial projects. Uses software such as SolidWorks, NX, Solid Edge and Inventor.	Good example of how mechanical engineers use CAD and new digital tools in daily engineering work. They also work with students. Design machines and industrial systems, create 3D CAD models and technical drawings, perform engineering calculations, and develop custom solutions for clients.
Gusella Bakker B.V.	Elst	Designs and manufactures hydraulic attachments and equipment for cranes and excavators.	Interesting because they combine mechanical design, CAD and actual manufacturing. We can ask how AI could improve their design and production process. Design hydraulic attachments for cranes and excavators, create production drawings, improve existing products, support manufacturing, and test new designs.
MPS Systems B.V.	Arnhem	Designs and manufactures complex printing presses for the global label industry.	Interesting because they combine mechanical engineering, automation, control systems and manufacturing. Design components and complete printing presses, develop mechanical systems, collaborate with automation and software engineers, and optimize machine performance and reliability.
Kelvion	Doetinchem	Designs and manufactures heat exchangers and industrial thermal equipment.	Their engineers use SolidWorks and 3D CAD, so they can give insight into how AI may support mechanical design and engineering calculations. Design heat exchangers and thermal systems, perform engineering calculations, create CAD models, select materials, and improve energy efficiency and product performance.
Von Gahlen	Zevenaar	Designs and builds specialised mechanical and	Interesting because they combine mechanical, electrical and software engineering, which makes AI applications especially relevant.

		mechatronic equipment.	Design specialized mechanical and mechatronic equipment, integrate mechanical, electrical and software components, develop prototypes, and test new solutions.
Royal SMIT Transformers	Nijmegen	Designs and manufactures large power transformers and electrical equipment.	Interesting because they use advanced 3D CAD and have a large engineering department where AI could be used for design, calculations and optimisation. Design transformers and electrical equipment, create 3D models and technical drawings, perform engineering calculations, optimize designs, and support manufacturing processes.

1.8 Internal & External Analysis: DESTEP, 7P and SWOT

Supporting analysis for the research proposal and PRD of the AI handbook for Dutch mechanical engineering and industrial product design SMEs (20–75 employees). DESTEP covers the external environment, 7P covers the handbook and its research platform as a product, and the SWOT brings both together.

1.8.1 External analysis DESTEP

Six external factors shaping how quickly the target audience will engage with AI, and the environment the handbook is published into.

FACTOR	DESCRIPTION
Demographic	An ageing workforce and a shortage of skilled mechanical/CAD engineers. Experienced designers are retiring and taking tacit design knowledge with them, while the younger engineers replacing them are more open to AI-supported tools — creating both urgency and an opening.
Economic	CBS reports AI adoption among Dutch companies (10+ employees) rising from 14% in 2023 to 22.7% in 2024, but adoption scales with size: ~17.8% at 10–19 employees versus 59.2% at 500+. Our 20–75 target group is under-represented. A typical SME implementation costs €5,000–10,000, so cost is a modest barrier; unclear ROI is the bigger one.
Social / Cultural	Engineering culture here is risk-averse and quality-driven, and AI output touching technical drawings or safety-relevant components raises legitimate trust concerns. Decision-makers rarely know what AI can and cannot do in a design context, producing a wait-and-see attitude rather than active resistance.
Technological	SolidWorks, Autodesk Inventor and Siemens NX are embedding generative design and AI-assisted modelling directly into their platforms, so firms need no separate AI

FACTOR	DESCRIPTION
	infrastructure. CBS data shows the largest barrier is experience, not cost: roughly 74–75% of non-adopters cite missing knowledge or skills.
Ecological	Generative design and AI-assisted optimisation can reduce material use and waste, aligning with EU sustainability and circular-economy pressure. This gives AI adoption a secondary justification for firms that must report on sustainability performance.
Political / Legal	The EU AI Act's high-risk obligations became enforceable on 2 August 2026, with penalties up to €35 million or 7% of global turnover, raising the stakes of adopting AI without understanding it. Meanwhile national and EU programmes (NL AI Coalition; the Digital Decade target of 90% of SMEs at basic digital intensity by 2030) are actively funding exactly this target group.

1.8.2 Internal analysis 7P

The marketing mix applied to the handbook and its supporting research platform as the product of this project.

P	ANALYSIS
Product	An English-language handbook, plus the research platform that produces it, explaining in practical terms how AI can support the design pipeline of mechanical engineering and product-design SMEs. Every claim is traceable to a source, so the product is curated and verified knowledge rather than plain information.
Price	No sales price — this is a research deliverable. The audience pays in time and attention; internally the cost is researcher time spent collecting, translating and verifying sources. The barrier to use stays low, but there is no revenue to sustain updates after the project ends.
Place	Distribution reuses existing wiki and website infrastructure (a project constraint), avoiding the cost of building a new channel. Reach therefore depends on how well that platform is already known to design engineers and SME owners.
Promotion	Awareness comes mainly from the research interviews: each interviewed company becomes a stakeholder with early exposure to the findings, supporting word-of-mouth via sector organisations and professional networks. There is no dedicated plan yet for SMEs outside the research sample.
People	A four-person student team collects and analyses sources and runs the platform. Primary users are non-technical researchers doing the gathering; secondary users are SME owners and design engineers. Success depends on the team's ability to

P	ANALYSIS
	interview credibly and judge source quality, since it is not yet domain-expert in mechanical engineering.
Process	A six-step workflow: collect sources, analyse, translate, generate a draft, human review, publish. Because translation and draft generation are both automated and error-prone, human review is the explicit quality gate before publication.
Physical evidence	The handbook pages themselves — consistent structure, APA references, HAN branding — and the published wiki presence. Consistency of tone, terminology and citation style signals rigor to a reader deciding whether to act on the content.

1.8.3 SWOT analysis

Strengths and weaknesses follow from the 7P analysis; opportunities and threats from DESTEP.

STRENGTHS (INTERNAL) • Academically grounded method: findings rest on document analysis and interviews, and every claim is traceable to a source. • Reuses existing wiki infrastructure, so no time goes into building a distribution channel. • Pre-defined quality criteria (understandable, practical, balanced, action-oriented, evidence-based) guide content from the start. • A four-person team lets research, translation, platform work and writing run in parallel.	WEAKNESSES (INTERNAL) • No budget or revenue model to maintain the handbook after the project ends. • The team is not yet domain-expert in mechanical engineering, risking missed nuance in technical sources. • Automated translation and AI-generated drafts carry a real risk of mistranslated terms and hallucinated claims. • Reliance on an agentic CLI rather than custom software limits control over workflow and output quality.
OPPORTUNITIES (EXTERNAL) • Dutch AI adoption is accelerating (14% → 22.7% in one year), creating a growing, receptive audience. • NL AI Coalition and EU Digital Decade programmes are funding exactly the SMEs we target. • AI is being built into the CAD platforms these firms already use, lowering the barrier the handbook must address. • The EU AI Act deadline creates urgency to understand AI responsibly — something the handbook can cover alongside opportunity.	THREATS (EXTERNAL) • The dominant barrier is missing skills (cited by ~three-quarters of non-adopters), which may also limit how much of the content lands without extra support. • Risk-averse engineering culture may cause readers to dismiss AI recommendations as impractical for safety-critical work. • AI capability evolves fast, so tool-specific references risk dating quickly. • Consultancies (€5,000–10,000 per SME project) and CAD vendors already compete for the same audience's attention.

1.9 Strategic implications

- Lead the handbook with verified, sourced examples rather than general claims, using the evidence-based method to counter a risk-averse audience.
- Turn the NL/EU funding programmes into a concrete recommendation section, since the main barrier is knowledge rather than budget.
- Treat human review as a formal QA gate, not an optional check, to offset the accuracy risk in translation and draft generation.
- Frame tool-specific content (SolidWorks, Inventor, NX) as examples of a broader trend, so the handbook dates more slowly.

2. AEL – Product requirements document

2.1 Problem and users

2.1.1 Problem

Researching AI applications for designers and engineers is time-consuming. To quantify this, we will first review AI features advertised by SolidWorks and other relevant engineering software vendors. We will then create a survey for HAN Mechanical Engineering and Automotive students, asking whether they use AI tools, how they use them, and how useful they believe these tools are. Where possible, the results of students who use AI will be compared with those who do not. This will help create a dataset showing which AI capabilities are being used and how effective they may be in engineering and CAD.

One limitation of this approach is that the test group is highly specialized, as all participants are students from similar technical study programs. This may introduce bias and means that the results may not represent professional engineers or the wider industry. This limitation could be mitigated by expanding the survey to other groups like employees of companies we interview or attempt to run the survey in social media.

The research team needs a platform that can efficiently collect, translate, analyse, and organize information into handbook pages.

2.1.2 Primary Users

- Student research team members
- Non-technical researchers who need to gather and analyse sources

2.1.3 Secondary Users

- SME owners
- Design engineers

These users will ultimately read the handbook to understand how AI can improve their design pipeline.

2.2 What must the user do

The platform should support the complete workflow from source collection to handbook publication.

1. Collect Sources	• Upload documents • Add website URLs • Add interview transcripts • Import Dutch and English sources
2. Analyse Sources	• Summarize documents • Extract key findings • Identify opportunities and risks • Detect information relevant to the research question

3. Translate Content	• Translate Dutch sources into English • Preserve technical terminology • Show original text alongside translations
4. Generate Handbook Draft	• Create handbook page drafts • Generate APA references • Suggest figures and examples
5. Human Review	• Edit generated prompt • Verify sources • Approve final handbook pages • Check wording against ISO 20607 4.4 comprehensibility principles
6. Publish	• Export handbook pages

2.3 Output Qualities

A handbook page produced by the platform should:

Output quality	Description
1. Be structured consistently	Each page follows a standard format, including: • Title • Introduction • Main findings • Practical examples • Recommendations • References
2. Be written in clear English	The content on the page is written in professional, easy-to-understand English suitable for engineers, designers, and SME owners. Terminology and wording follow the comprehensibility principles described in ISO 20607:2019, Clause 4.4 (Comprehensible terminology and wording).
3. Be traceable to sources	Every key claim, recommendation, and finding is linked to a supporting source or interview.
4. Maintain a consistent style	Pages use the same tone, terminology, formatting, and citation style.
5. include citations	Sources are referenced according to APA 7th edition.
6. Protect interview participants (<i>Response to lecturer feedback (week 2), question 2.)</i>	No handbook page names an interviewee, their employer or any detail that identifies them, unless that company has given written consent to be named.

2.4 Known constraints

Requirements for literature / sources

- Literature is no more than 4 years old.
- Sources are academic or professional (no Wikipedia or blogs without an author).
- Language: Dutch and/or English.
- Sources originate from recognized publishers, professional journals, government bodies, or reputable knowledge organizations.

Agentic CLI

- The platform must be operated through an agentic CLI instead of custom-coded software.
- Design consequence: the platform is a set of instructions and file contracts, not an application. Behaviour changes by editing an instruction rather than rewriting code, so a team without developers can maintain it; in exchange, every part performed by a model needs a human checkpoint before publication. Why this holds for our three language tasks is set out in blueprint 1.7.

Existing Infrastructure

- The project must reuse existing wiki and website infrastructure.

Response to lecture feedback (week 2), question 1.

- Non-coder operation
 - The platform must be runnable by a team member who does not program.
 - *Design consequence: no part of the pipeline may require writing, compiling or maintaining code. Each part is defined by written instructions and by the shape of the file it hands on, so a new team member can run it by following the instructions.*
- Manual fallback
 - Every step of the pipeline must also be doable by a person.
 - *Design consequence: each of the six parts (Collect, Appraise, Translate, Draft, Check, Publish) is defined by its input file and its output file, not by the tool that produces it. If the model service is unavailable, too expensive or gives unusable output, a team member can produce the same file by hand and the pipeline continues.*

Response to lecture feedback (week 2), question 2.

Handling of interview material

- Interview transcripts and recordings are the team's own field material and contain personal data: the name of the interviewee, their role, their employer and their opinions about their own work.
- Raw, unredacted interview material must never be sent to an external model service. Only material from which person- and company-identifying details have been removed by a team member may leave the platform.
- Design consequence: the platform needs a single, enforced exit point rather than letting each part call an external service, and a human redaction step that a source must pass before it can be processed. This is implemented as the Gateway (blueprint, Decision 5 and 1.4).
- Rationale: (a) the transcripts contain personal data and the team has no data-processing agreement with any model provider, so we treat sending them as a risk we do not take; (b) the SMEs in 1.5 grant interviews on the understanding that their material is not passed to third parties; (c) the handbook's own chapter template requires field material to be "anonymised or published with consent only".

Response to lecturer feedback (week 2), question 3.

Model service behind the Gateway

- The platform calls exactly one model service at a time, and which one is a team decision recorded here, not an implementation detail chosen inside a part.
- Permitted at this stage: **[number del service]**. Any change requires a new PRD version.
- Where a task can be done by a European or open-weight model of acceptable quality, we prefer it over a US-hosted service.
- *Design consequence: because only Gateway talks to the outside world (blueprint, Decision 5), switching model service is a change in one place, not in six parts. This keeps the option of moving to a European or locally hosted model open rather than closing it by accident.*

Rationale: our own handbook chapter on geopolitics and vendor models advises SMEs to know which provider they depend on and to keep a fallback. The platform follows the same rule it publishes. The observation that drove this requirement is in the research proposal (AIBS 1.1), which notes that AI features in CAD software concentrate dependence on mainly U.S.-based providers; this constraint turns that observation into a commitment about our own platform.

2.5 Source material

Many useful sources for Dutch SMEs are written in Dutch, including:

- CBS reports

- Chamber of Commerce publications
- Sector organizations
- Regional news articles
- Interview transcripts
- Prohibited inputs: raw interview transcripts or recordings that have not been redacted; any source containing personal data of identifiable individuals that the team has not anonymised; material a company has asked us not to reuse.

2.6 Out of scope (for now)

The platform will NOT:

- Build custom AI models
- Replace human decision making
- Conduct interviews itself
- Generate engineering designs
- Integrate directly into CAD software

Reason for excluding custom AI models: existing generative AI tools are already sufficient for the summarisation, translation, and extraction tasks this platform requires, and training custom models would require data, budget, and time disproportionate to a project of this scope. Off-the-shelf tools are therefore used as-is; this may be reconsidered in future versions.

2.7 Open questions and assumptions

2.7.1 Open questions

- Has any comparable mechanical-engineering/product-design SME already implemented AI in its design pipeline? How will such a case, if it exists, be found and verified?
- What counts as a "successful" reduction in research time is there a target percentage or number, or just "faster than the 37-minute baseline"?
- Who maintains or updates the handbook after the project ends, given there's currently no budget or owner for that (flagged as a weakness in the 7P/SWOT work)?

2.7.2 Assumptions

- Decision-makers at target SMEs currently lack sufficient knowledge of AI's capabilities not yet tested, currently just asserted in AIBS 1.2.
- The innovation gap Draghi describes between the EU and US/China applies meaningfully at the level of individual CAD-tool AI features, not just macro R&D investment.
- SME decision-makers will read and act on a written handbook, rather than needing hands-on consultancy support to change behaviour.
- Design consequence: the platform is a set of instructions and file contracts, not an application. Behaviour changes by editing an instruction, so a team without developers can maintain it; in exchange, every part that a model performs needs a human checkpoint before publication.

3.The Knowledge Architecture

3.1 What the platform has to know

What the platform reads, and where each item comes from:

What	Where it comes from
Raw sources: uploaded documents, website URLs, interview transcripts	Supplied by a team member at Collect (PRD 2.2, step 1).
source.md → appraised.md → translated.md → draft.md → page.md	The pipeline's own file contract; each role writes one and the next role reads it (blueprint Decision 3, 1.2.1).
The shared context file (week 2)	Team-authored: PRD 2.4 sourcing rules, the AIBS 1.4 / PRD 2.3 quality criteria, the ISO 20607 wording principles, and the team's terminology list. Consolidated per blueprint Decision 4; once consolidated, every part reads it by default.
The chapter template	Fixed structure — Title, Introduction, Main findings, Practical examples, Recommendations, References (PRD 2.3). Read by Draft when it assembles a page.
Style and branding conventions	APA 7 formatting and HAN branding, carried over from the existing wiki/website infrastructure the project reuses (PRD 2.4, "Existing Infrastructure"). Read by Draft and Check.

3.2 Where it lives, and how it is found again

Our choice: compile ahead of time, addressed by an explicit id — not a query-time lookup over an index.

Draft does not search the whole source pool when it runs. "Search" already happened earlier: Collect and Appraise gather and quality-check sources into a pool of appraised.md / translated.md files, each carrying a source_id (blueprint 1.3). For a given handbook chapter, a team member picks which source_ids belong in it. Draft then opens exactly those files by their id and compiles them into one draft, with every claim tagged with the source_id it came from (blueprint 1.2.1). There is no moment where a model searches across the whole corpus by similarity; retrieval is a person's selection followed by a deterministic file lookup.

The path from a question to the passage that answers it:

Chapter question	Source pool (ids)	Person selects ids	Draft opens those files	draft.md (claims tagged with source_id)	Check / Publish
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3.3 Why this, and not the alternatives

Alternative weighed: query-time semantic retrieval build a vector index over all appraised/translated sources and let the model automatically retrieve the top-matching passages whenever Draft runs for a chapter (a classic RAG setup).

3.3.1 Why we did not choose it

- PRD promise served: PRD 2.3 requires every handbook page to "be traceable to sources", with every claim linked to a supporting source. Human-selected, id-addressed sources make that link exact and auditable. A similarity search can surface a passage that reads as related without being the source a claim should rest on — weaker traceability, not stronger.
- PRD 2.4 constraints: the platform must be non-coder operable and run on an agentic CLI rather than custom software. Building and maintaining a vector index is infrastructure and expertise a four-person team with no developer does not have, and the PRD rules this out by design. Deterministic file lookup needs nothing beyond the file contract already decided (blueprint Decision 3).
- Manual fallback (PRD 2.4): every pipeline step must also be doable by hand. A team member can open the right file by its id if the CLI is unavailable; a vector index cannot be operated manually.

3.3.2 What would make us switch

Two conditions: (1) the source pool grows so large that a person can no longer reasonably identify which source_ids belong in each chapter, so manual selection becomes the bottleneck rather than a quality safeguard.

(2) a later version of the platform must answer live questions from readers, not only produce static handbook pages at that point a query-time retrieval layer would earn the complexity it costs today

3.4 The line, kept

In plain terms: interview transcripts are the most sensitive material we hold they carry someone's name, employer and personal opinions. The blueprint's rule (1.4) is that this raw material may never be sent to an external AI model. Below is exactly where that raw material sits, and exactly what mechanism stops it from being sent out — not a promise on paper, but a checkpoint built into the platform.

3.4.1 Where it is stored

Nothing special or hidden: a raw interview transcript is saved as an ordinary source.md file, in the very same place every other source lands after Collect (PRD 2.2). It is not quarantined in a

separate secret location. What keeps it protected is not where it sits, but that it is not allowed to move any further through the pipeline until a person has cleared it described step by step below.

3.4.2 What stops it from reaching a model step by step

1	When Collect saves an interview transcript, it is automatically labelled <code>source_type = "interview"</code> and <code>review_status = "pending"</code> (blueprint 1.3). This labelling happens for every source, not as an extra step.
2	Every source that is ever going to reach a model must pass through the Gateway the one and only part of the platform allowed to contact an external model service (blueprint Decision 5). No other role (Collect, Appraise, Translate, Draft, Check, Publish) is permitted to call out on its own.
3	The Gateway performs one fixed check before anything else happens: if <code>source_type = "interview"</code> and <code>review_status</code> is not "redacted", it refuses to send the source. This check runs before the model is even involved — the model never gets the chance to see the raw text and decide what to do with it.
4	The only way <code>review_status</code> can change from "pending" to "redacted" is a team member manually removing the interviewee's name, employer and any other identifying detail from the transcript. This is a human action. No part of the pipeline, automated or not, can perform this step or skip it.
5	Only once <code>review_status = "redacted"</code> does the Gateway allow that (now-anonymised) content through to the model service.

1. source.md saved (interview, status: pending)	2. every source must pass the Gateway	3. Gateway checks: type=interview + not redacted → refuse	4. team member redacts by hand → status: redacted	5. Gateway allows through	model service
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